

COA Question Bank for Mid SEM

1. Explain Von-Neumann (8086) Architecture in details. Explain all its functional units.
2. Explain basic computer architecture (IAS). How further, upcoming architectures were evolved further.
3. Explain various modes of retrieval of instructions and data from the memory.
4. What are the different addressing modes (8086)? Explain all in detail.
5. Explain below instructions with its addressing modes:
 - 5.1 MOV DX, 2500H
 - 5.2 MOV AL, [1000H]
 - 5.3 MOV SI, [1010H]
 - 5.4 MOV [BX], CL
 - 5.5 MOV AX, [SI+10H]
 - 5.6 MOV [BP+DI], CX
 - 5.7 MOV BYTE PTR [DI], 25H
6. Explain bus structure and its types in detail.
7. Multiply the following using Booth's algorithm (Multiplicand= +11 Multiplier= -6), (Multiplicand=6 and Multiplier = 4)
8. Represent below numbers in 32 bit single precision and 64 bit double precision format. $(-65.23)_{10}$, $(124.16)_{10}$
9. Multiply following operands using recoded multiplier technique $(+12 \times -3)$, $(+16 \times -7)$
10. Multiply the following pair of signed 2's complement numbers using bit-pair- recoding Booth's Algorithm Multiplicand: 1 1 0 0 0 1 Multiplier: 1 0 1 1 0 1
11. Perform division of following numbers using Restoring Division Algorithm: Dividend=1000 Divisor=0001, Dividend=23 Divisor=5
12. Perform division of following numbers using Non- Restoring Division Algorithm: Dividend=1010 Divisor=0011, Dividend=15 Divisor=3

ALL THE BEST